MODULE 8: PROJECT PART 3

GitHub Repository Link: https://github.com/devcnx/csd430

styles.css

```
/*
    styles.css - External CSS for Module 4 assignment
    Author: Brittaney Perry-Morgan
    Description: Styles for the JavaBean movie data display.
 */

/* ==========================
   1. GLOBAL STYLES
   ========================== */

*,
html,
body {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Arial, sans-serif;
    background: linear-gradient(120deg, #f8fafc 0%, #e0e7ef 100%);
    margin: 0;
    padding: 20px;
    min-height: 100vh;
}

/* ==========================
   2. LAYOUT & CONTAINER
   ========================== */

.container,
.top-title {
    max-width: 900px;
    margin: 0 auto;
    padding: 20px;
}
```

```
.container {
    background: #fff;
    border-radius: 12px;
    box-shadow: 0 4px 24px rgba(0, 0, 0, 0.08);
    margin-bottom: 20px;
}

.top-title {
    text-align: center;
    background: linear-gradient(135deg, #2a3a5e 0%, #1a2540 100%);
    color: #fff;
}

.top-title h1 {
    color: #fff;
    margin: 0 0 10px 0;
}

.top-title h2 {
    color: #e0e7ef;
    margin: 0;
    font-weight: 400;
}

.top-title h2 em {
    font-style: normal;
    color: #a3d9ff;
}

/* ========================
   3. TYPOGRAPHY
   ======================== */

h1,
h2,
h3 {
    color: #2a3a5e;
    margin: 20px 0 15px 0;
    text-transform: uppercase;
}
```

```
h1 {
    font-weight: 200;
    font-size: 1.8rem;
}

h2 {
    font-weight: 400;
    font-size: 1.4rem;
}

h3 {
    font-weight: 600;
    font-size: 1.1rem;
    color: #1a2540;
    border-bottom: 2px solid #e3eaf6;
    padding-bottom: 8px;
}

p.description {
    color: #ffffff;
    line-height: 1.7;
    margin: 15px auto;
    padding: 0 20px;
}

/* ========================
   4. LISTS
   ======================== */

.field-descriptions {
    list-style: none;
    padding: 0 20px;
}

.field-descriptions li {
    padding: 10px 15px;
    margin: 8px 0;
    background: #f1f5fa;
    border-radius: 6px;
    border-left: 4px solid #2a3a5e;
    color: #444;
```

```
}

/* ========================
   5. FORMS
   ========================= */

.form-table {
    width: 100%;
    border-collapse: collapse;
    margin: 20px;
    background: #fff;
}

.form-table tr {
    background: #fff;
}

.form-table td {
    border: 1px solid #e3eaf6;
    padding: 12px 15px;
    vertical-align: top;
}

.form-table td:first-child {
    background: #f9fbfd;
    width: 25%;
    font-weight: 600;
    color: #1a2540;
    vertical-align: middle;
}

.form-table td:last-child {
    width: 75%;
}

.form-table label {
    display: block;
    margin-bottom: 5px;
    font-size: 0.9rem;
    color: #333;
}
```

```
.form-table input[type="text"],
.form-table input[type="email"],
.form-table input[type="tel"],
.form-table input[type="number"],
.form-table input[type="date"],
.form-table select,
.form-table textarea {
    width: 100%;
    padding: 10px 12px;
    border: 1px solid #ccc;
    border-radius: 6px;
    font-size: 1rem;
    font-family: 'Segoe UI', Arial, sans-serif;
    box-sizing: border-box;
}

.form-table select {
    background: #fff;
    cursor: pointer;
}

.form-table textarea {
    resize: vertical;
    min-height: 80px;
}

.form-table input[type="checkbox"] {
    width: auto;
    margin-right: 8px;
}

/* ========================
   6. NAVIGATION
   ======================== */

nav ul {
    list-style: none;
    padding: 0 20px;
}
```

```
nav ul li {
    margin: 10px 0;
}

nav ul li a {
    color: #2a3a5e;
    text-decoration: none;
    font-weight: 600;
    padding: 8px 12px;
    background: #f1f5fa;
    border-radius: 6px;
    transition: all 0.2s ease;
}

nav ul li a:hover {
    background: #2a3a5e;
    color: #fff;
}

.nav-desc {
    display: block;
    color: #666;
    font-size: 0.85rem;
    margin-top: 8px;
    font-weight: 400;
}

/* ========================
   6. FORM ACTIONS (for button links)
   ========================= */

.form-actions {
    margin: 30px 20px;
    padding: 15px;
    background: #f9fbfd;
    border-radius: 8px;
    text-align: center;
}

.btn-submit,
.btn-reset {
```

```
    padding: 12px 24px;
    margin: 0 10px;
    font-size: 1rem;
    font-weight: 600;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    transition: all 0.2s ease;
    text-decoration: none;
    display: inline-block;
}

.btn-submit {
    background: #2a3a5e;
    color: #fff;
}

.btn-submit:hover {
    background: #1a2540;
    transform: translateY(-1px);
    box-shadow: 0 4px 12px rgba(42, 58, 94, 0.3);
}

.btn-reset {
    background: #e3eaf6;
    color: #2a3a5e;
}

.btn-reset:hover {
    background: #cdd8eb;
}

/* ========================
   7. FORM ACTIONS (for button links)
   ========================= */

.form-actions {
    margin: 30px 20px;
    padding: 15px;
    background: #f9fbfd;
    border-radius: 8px;
```

```
    text-align: center;
}

.btn-submit,
.btn-reset {
    padding: 12px 24px;
    margin: 0 10px;
    font-size: 1rem;
    font-weight: 600;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    transition: all 0.2s ease;
    text-decoration: none;
    display: inline-block;
}

.btn-submit {
    background: #2a3a5e;
    color: #fff;
}

.btn-submit:hover {
    background: #1a2540;
    transform: translateY(-1px);
    box-shadow: 0 4px 12px rgba(42, 58, 94, 0.3);
}

.btn-reset {
    background: #e3eaf6;
    color: #2a3a5e;
}

.btn-reset:hover {
    background: #cdd8eb;
}

/* ==========================
   8. TABLES
   ========================== */
```

```
table {
    width: 100%;
    border-collapse: collapse;
    margin: 20px;
    background: #fff;
}

thead {
    background: #2a3a5e;
    color: #fff;
}

th,
td {
    border: 1px solid #e3eaf6;
    padding: 12px 15px;
    text-align: left;
}

th {
    font-weight: 600;
    font-size: 0.85rem;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}

td {
    font-size: 0.95rem;
    color: #333;
    vertical-align: top;
}

tr:nth-child(even) {
    background: #f9fbfd;
}

tr:hover {
    background: #e8f0ff;
    transition: background 0.2s ease;
}
```

```
/* ========================
   9. FOOTER
   ======================== */

footer {
    text-align: center;
    padding: 20px;
    color: #666;
    font-size: 0.9rem;
}

footer p {
    margin: 0;
}

/* ========================
   10. RESPONSIVE DESIGN
   ======================== */

@media (max-width: 768px) {
    body {
        padding: 10px;
    }

    .container,
    .top-title {
        padding: 15px;
    }

    h1 {
        font-size: 1.4rem;
    }

    h2 {
        font-size: 1.1rem;
    }

    h3 {
        font-size: 1rem;
    }
```

```
.form-table {
    margin: 10px 0;
    font-size: 0.85rem;
}

.form-table td {
    padding: 8px 10px;
}

.form-table td:first-child {
    font-size: 0.85rem;
}

.form-table input[type="text"],
.form-table input[type="email"],
.form-table input[type="tel"],
.form-table input[type="number"],
.form-table input[type="date"],
.form-table select,
.form-table textarea {
    font-size: 0.9rem;
    padding: 8px;
}

table {
    margin: 10px 0;
    font-size: 0.85rem;
}

th,
td {
    padding: 8px 10px;
}

p.description {
    padding: 0 10px;
}

.field-descriptions {
    padding: 0 10px;
}
```

```
    .field-descriptions li {
        padding: 8px 12px;
    }

    nav ul {
        padding: 0 10px;
    }

    .form-actions {
        margin: 20px 10px;
        padding: 10px;
    }

    .btn-submit,
    .btn-reset {
        padding: 10px 16px;
        font-size: 0.9rem;
        margin: 5px;
    }
}
```

MovieBean.java

```
/**
 * MovieBean.java
 *
 * CSD430 Server-Side Development
 * Module 8 Project Part 3: CRUD-UPDATE Operations
 *
 * Author: Brittaney Perry-Morgan
 *
 * Description: JavaBean for managing movie database operations using JDBC.
 *              Provides methods for database connectivity, movie retrieval,
 *              insertion, and updates. Used by JSP pages for CRUD operations.
 *
 * Database: CSD430
 * Table: brittaney_movies_data
 *
 * Fields:
 *   - movie_id (INT, Primary Key, Auto-increment)
 *   - title (VARCHAR)
 *   - genre (VARCHAR)
```

```
 *   - release_year (INT)
 *   - rating (VARCHAR)
 *   - director (VARCHAR)
 */

package csd430;

import java.io. *;
import java.sql. *;
import java.util.ArrayList;
import java.util.List;

/**
 * JavaBean class for movie database operations.
 * Handles JDBC connectivity and CRUD operations for the movie database.
 *
 * <p>This bean follows the JavaBeans convention with a no-argument constructor
 * and getter/setter methods. Database credentials are loaded from environment
 * variables or a .env file.</p>
 *
 * @author Brittaney Perry-Morgan
 * @version 1.0
 * @since Module 8 Project Part 3
 */
public class MovieBean {

    // Database connection configuration
    private static String DB_URL;
    private static String DB_USER;
    private static String DB_PASSWORD;

    // Static initialization block - loads database credentials on class load
    static {
        loadEnvironmentVariables();
    }

    /**
     * Loads database connection credentials from environment variables or .env file.
     * Falls back to default values if not found.
     *
     * <p>Environment variables take precedence:
```

```
 * <ul>
 *   <li>DB_URL - JDBC connection URL</li>
 *   <li>DB_USER - Database username</li>
 *   <li>DB_PASSWORD - Database password</li>
 * </ul></p>
 *
 * <p>Default fallbacks:
 * <ul>
 *   <li>DB_URL: jdbc:mysql://localhost:3306/CSD430</li>
 *   <li>DB_USER: student1</li>
 *   <li>DB_PASSWORD: pass</li>
 * </ul></p>
 */
private static void loadEnvironmentVariables() {
    DB_URL = System.getenv("DB_URL");
    DB_USER = System.getenv("DB_USER");
    DB_PASSWORD = System.getenv("DB_PASSWORD");

    if (DB_URL == null || DB_USER == null || DB_PASSWORD == null) {
        String envFile = System.getProperty("user.dir") + "/.env";
        if (new File(envFile).exists()) {
            try (BufferedReader reader = new BufferedReader(new FileReader(envFile))) {
                String line;
                while ((line = reader.readLine()) != null) {
                    if (!line.trim().isEmpty() && !line.startsWith("#")) {
                        String[] parts = line.split("=", 2);
                        if (parts.length == 2) {
                            String key = parts[0].trim();
                            String value = parts[1].trim();
                            switch (key) {
                                case "DB_URL": DB_URL = value; break;
                                case "DB_USER": DB_USER = value; break;
                                case "DB_PASSWORD": DB_PASSWORD = value; break;
                            }
                        }
                    }
                }
            } catch (IOException e) {
                System.err.println("Error loading .env file: " + e.getMessage());
            }
        }
```

```
    }

    // Default fallback values
    if (DB_URL == null) DB_URL = "jdbc:mysql://localhost:3306/CSD430";
    if (DB_USER == null) DB_USER = "student1";
    if (DB_PASSWORD == null) DB_PASSWORD = "pass";
}

// Instance variables
private Connection connection;
private List<Integer> movieIds;
private ResultSet resultSet;
private String errorMessage;

/**
 * Default constructor - initializes the movie ID list and error message.
 * Required for JavaBeans compatibility.
 */
public MovieBean() {
    movieIds = new ArrayList<Integer>();
    errorMessage = null;
}

/**
 * Establishes a connection to the MySQL database using loaded credentials.
 * Loads the MySQL JDBC driver and creates a connection.
 *
 * @return true if connection successful, false otherwise
 */
public boolean connect() {
    try {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            Class.forName("com.mysql.jdbc.Driver");
        }
        connection = DriverManager.getConnection(DB_URL, DB_USER, DB_PASSWORD);
        return true;
    } catch (ClassNotFoundException e) {
        errorMessage = "MySQL Driver not found: " + e.getMessage();
        return false;
```

```
    } catch (SQLException e) {
        errorMessage = "Database connection error: " + e.getMessage();
        return false;
    }
}

/**
 * Closes the database connection and any open result sets.
 * Should be called when database operations are complete.
 */
public void disconnect() {
    try {
        if (resultSet != null) {
            resultSet.close();
        }
        if (connection != null) {
            connection.close();
        }
    } catch (SQLException e) {
        errorMessage = "Error closing connection: " + e.getMessage();
    }
}

/**
 * Retrieves all movie IDs from the database and stores them in the movieIds list.
 * Results are ordered by movie_id.
 *
 * @return true if successful, false if an error occurred
 */
public boolean getAllMovieIds() {
    if (connection == null) {
        if (!connect()) {
            return false;
        }
    }

    try {
        String query = "SELECT movie_id FROM brittaney_movies_data ORDER BY movie_id";
        PreparedStatement stmt = connection.prepareStatement(query);
        resultSet = stmt.executeQuery();
```

```
        movieIds.clear();
        while (resultSet.next()) {
            movieIds.add(resultSet.getInt("movie_id"));
        }
        resultSet.close();
        stmt.close();
        return true;
    } catch (SQLException e) {
        errorMessage = "Error fetching movie IDs: " + e.getMessage();
        return false;
    }
}

/**
 * Retrieves a single movie by its ID and populates the result set.
 * Used for fetching movie details for editing.
 *
 * @param movieId the primary key of the movie to retrieve
 * @return true if movie found, false if not found or error occurred
 */
public boolean getMovieById(int movieId) {
    if (connection == null) {
        if (!connect()) {
            return false;
        }
    }

    try {
        String query = "SELECT * FROM brittaney_movies_data WHERE movie_id = ?";
        PreparedStatement stmt = connection.prepareStatement(query);
        stmt.setInt(1, movieId);
        resultSet = stmt.executeQuery();

        if (!resultSet.next()) {
            errorMessage = "Movie not found";
            resultSet.close();
            stmt.close();
            return false;
        }
        return true;
    } catch (SQLException e) {
```

```
        errorMessage = "Error fetching movie: " + e.getMessage();
        return false;
    }
}

/**
 * Retrieves all movies from the database in a result set.
 * Results are ordered by movie_id.
 *
 * @return true if successful, false if an error occurred
 */
public boolean getAllMovies() {
    if (connection == null) {
        if (!connect()) {
            return false;
        }
    }

    try {
        String query = "SELECT * FROM brittaney_movies_data ORDER BY movie_id";
        PreparedStatement stmt = connection.prepareStatement(query);
        resultSet = stmt.executeQuery();
        return true;
    } catch (SQLException e) {
        errorMessage = "Error fetching movies: " + e.getMessage();
        return false;
    }
}

/**
 * Inserts a new movie record into the database.
 * The movie_id is auto-generated by the database.
 *
 * @param title the movie title
 * @param genre the movie genre
 * @param releaseYear the year of release
 * @param rating the MPAA rating
 * @param director the director's name
 * @return true if insert successful, false otherwise
 */
public boolean insertMovie(String title, String genre, int releaseYear, String rating, String director) {
```

```
    if (connection == null) {
        if (!connect()) {
            return false;
        }
    }

    try {
        String query = "INSERT INTO brittaney_movies_data (title, genre, release_year, rating, director) VALUES (?, ?, ?, ?, ?)";
        PreparedStatement stmt = connection.prepareStatement(query);
        stmt.setString(1, title);
        stmt.setString(2, genre);
        stmt.setInt(3, releaseYear);
        stmt.setString(4, rating);
        stmt.setString(5, director);

        int rowsAffected = stmt.executeUpdate();
        stmt.close();
        return rowsAffected > 0;
    } catch (SQLException e) {
        errorMessage = "Error inserting movie: " + e.getMessage();
        return false;
    }
}

 /**
  * Updates an existing movie record in the database.
  * All fields except movie_id can be modified.
  *
  * @param movieId the primary key of the movie to update
  * @param title the new movie title
  * @param genre the new movie genre
  * @param releaseYear the new release year
  * @param rating the new MPAA rating
  * @param director the new director's name
  * @return true if update successful, false otherwise
  */
 public boolean updateMovie(int movieId, String title, String genre, int releaseYear, String rating, String director) {
    if (connection == null) {
        if (!connect()) {
            return false;
        }
```

```
        }

        try {
            String query = "UPDATE brittaney_movies_data SET title = ?, genre = ?, release_year = ?, rating = ?, director = ? WHERE
movie_id = ?";
            PreparedStatement stmt = connection.prepareStatement(query);
            stmt.setString(1, title);
            stmt.setString(2, genre);
            stmt.setInt(3, releaseYear);
            stmt.setString(4, rating);
            stmt.setString(5, director);
            stmt.setInt(6, movieId);

            int rowsAffected = stmt.executeUpdate();
            stmt.close();
            return rowsAffected > 0;
        } catch (SQLException e) {
            errorMessage = "Error updating movie: " + e.getMessage();
            return false;
        }
    }

    /**
     * Returns the list of movie IDs from the last getAllMovieIds() call.
     *
     * @return List of movie IDs
     */
    public List<Integer> getMovieIds() {
        return movieIds;
    }

    /**
     * Returns the movie_id from the current result set position.
     *
     * @return the movie ID, or 0 if error
     */
    public int getMovieId() {
        try {
            return resultSet.getInt("movie_id");
        } catch (SQLException e) {
            return 0;
```

```
    }
}

/**
 * Returns the title from the current result set position.
 *
 * @return the movie title, or empty string if error
 */
public String getTitle() {
    try {
        return resultSet.getString("title");
    } catch (SQLException e) {
        return "";
    }
}

/**
 * Returns the genre from the current result set position.
 *
 * @return the movie genre, or empty string if error
 */
public String getGenre() {
    try {
        return resultSet.getString("genre");
    } catch (SQLException e) {
        return "";
    }
}

/**
 * Returns the release year from the current result set position.
 *
 * @return the release year, or 0 if error
 */
public int getReleaseYear() {
    try {
        return resultSet.getInt("release_year");
    } catch (SQLException e) {
        return 0;
    }
}
```

```java
/**
 * Returns the rating from the current result set position.
 *
 * @return the MPAA rating, or empty string if error
 */
public String getRating() {
    try {
        return resultSet.getString("rating");
    } catch (SQLException e) {
        return "";
    }
}

/**
 * Returns the director from the current result set position.
 *
 * @return the director name, or empty string if error
 */
public String getDirector() {
    try {
        return resultSet.getString("director");
    } catch (SQLException e) {
        return "";
    }
}

/**
 * Returns the last error message from database operations.
 *
 * @return error message string, or null if no error
 */
public String getErrorMessage() {
    return errorMessage;
}

/**
 * Advances the result set to the next row.
 * Used for iterating through query results.
 *
 * @return true if there is a next row, false if end of results
```

```
     */
    public boolean next() {
        try {
            return resultSet.next();
        } catch (SQLException e) {
            errorMessage = "Error navigating result set: " + e.getMessage();
            return false;
        }
    }
}
```

movieDisplay.jsp

```
<%--
    CSD430 Server-Side Development

    Module 8 Project Part 3: CRUD-UPDATE Operations
    Author: Brittaney Perry-Morgan

    Description: Displays all movies with edit links for UPDATE operations.
    Users can select a movie to edit its details.

    movieDisplay.jsp - Movie List Display Page
--%>
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="csd430.MovieBean" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Movie List - CSD430</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 8 Project Part 3</h1>
        <h2>Movie Database - <em>UPDATE Operation</em></h2>
        <p class="description">Select a movie to edit its details.</p>
    </div>

    <div class="container">
<%
```

```
    MovieBean movieBean = new MovieBean();
    boolean connected = movieBean.connect();
    int recordCount = 0;

    if (!connected) {
%>
    <div class="error-message">
        <p>Database Connection Error:</p>
        <p><%= movieBean.getErrorMessage() %></p>
    </div>
<%
    } else {
        boolean success = movieBean.getAllMovies();

        if (!success) {
%>
    <div class="error-message">
        <p>Error retrieving movie data:</p>
        <p><%= movieBean.getErrorMessage() %></p>
    </div>
<%
        } else {
            // Count records
            while (movieBean.next()) {
                recordCount++;
            }
            movieBean.disconnect();
%>

    <p class="record-count">Total Records: <%= recordCount %></p>

    <table>
        <thead>
            <tr>
                <th>Movie ID</th>
                <th>Title</th>
                <th>Genre</th>
                <th>Release Year</th>
                <th>Rating</th>
                <th>Director</th>
                <th>Actions</th>
```

```
            </tr>
        </thead>
        <tbody>
<%
            // Re-query for display
            movieBean.connect();
            movieBean.getAllMovies();
            while (movieBean.next()) {
%>
            <tr>
                <td><%= movieBean.getMovieId() %></td>
                <td><%= movieBean.getTitle() %></td>
                <td><%= movieBean.getGenre() %></td>
                <td><%= movieBean.getReleaseYear() %></td>
                <td><%= movieBean.getRating() %></td>
                <td><%= movieBean.getDirector() %></td>
                <td>
                    <a href="movieEdit.jsp?id=<%= movieBean.getMovieId() %>" class="btn-submit">Edit</a>
                </td>
            </tr>
<%
            }
            movieBean.disconnect();
%>
        </tbody>
    </table>

    <h3>Field Descriptions</h3>
    <ul class="field-descriptions">
        <li><strong>Movie ID:</strong> Unique identifier (Primary Key)</li>
        <li><strong>Title:</strong> The name of the movie</li>
        <li><strong>Genre:</strong> The category of the movie</li>
        <li><strong>Release Year:</strong> Year the movie was released</li>
        <li><strong>Rating:</strong> MPAA rating</li>
        <li><strong>Director:</strong> Director of the movie</li>
    </ul>
<%
        }
    }
%>
```

```
        <div class="form-actions">
            <a href="/module_07_project_2/movieForm.jsp" class="btn-submit">Add New Movie</a>
            <a href="index.jsp" class="btn-reset">Back to Course Index</a>
        </div>
    </div>

    <footer>
        <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
</body>
</html>
```

movieEdit.jsp

```
<%--
    CSD430 Server-Side Development

    Module 8 Project Part 3: CRUD-UPDATE Operations
    Author: Brittaney Perry-Morgan

    Description: Form to edit an existing movie record. Pre-populated
    with current data from the database.

    movieEdit.jsp - Edit Movie Form Page
--%>
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="csd430.MovieBean" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Edit Movie - CSD430</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 8 Project Part 3</h1>
        <h2>Movie Database - <em>UPDATE Operation</em></h2>
        <p class="description">Edit the movie details below.</p>
    </div>

    <div class="container">
```

```
<%
    String idParam = request.getParameter("id");
    MovieBean movieBean = new MovieBean();
    String errorMessage = null;

    // Variables to store movie data
    int movieId = 0;
    String title = "";
    String genre = "";
    int releaseYear = 0;
    String rating = "";
    String director = "";

    if (idParam == null || idParam.isEmpty()) {
        errorMessage = "No movie ID provided. Please select a movie to edit.";
    } else {
        try {
            movieId = Integer.parseInt(idParam);
            boolean connected = movieBean.connect();

            if (!connected) {
                errorMessage = "Database connection error: " + movieBean.getErrorMessage();
            } else {
                boolean found = movieBean.getMovieById(movieId);

                if (!found) {
                    errorMessage = "Movie not found with ID: " + movieId;
                    movieBean.disconnect();
                } else {
                    // Store values before disconnecting
                    title = movieBean.getTitle();
                    genre = movieBean.getGenre();
                    releaseYear = movieBean.getReleaseYear();
                    rating = movieBean.getRating();
                    director = movieBean.getDirector();
                    movieBean.disconnect();
                }
            }
        } catch (NumberFormatException e) {
            errorMessage = "Invalid movie ID: " + idParam;
        }
```

```
    }

    // Display error message if any
    if (errorMessage != null) {
%>
    <div class="error-message">
        <p><%= errorMessage %></p>
    </div>
    <div class="form-actions">
        <a href="movieSelection.jsp" class="btn-submit">Select Movie</a>
        <a href="index.jsp" class="btn-reset">Back to Course Index</a>
    </div>
<%
    } else {
%>

    <h3>Edit Movie</h3>
    <p class="description">Movie ID is the primary key and cannot be modified.</p>

    <form action="movieUpdate.jsp" method="post">
        <input type="hidden" name="movieId" value="<%= movieId %>">

        <table class="form-table">
            <tr>
                <td><label for="movieIdDisplay">Movie ID:</label></td>
                <td><input type="text" id="movieIdDisplay" value="<%= movieId %>" readonly class="readonly-field"></td>
            </tr>
            <tr>
                <td><label for="title">Movie Title:</label></td>
                <td><input type="text" id="title" name="title" required value="<%= title %>"></td>
            </tr>
            <tr>
                <td><label for="genre">Genre:</label></td>
                <td>
                    <select id="genre" name="genre" required>
                        <option value="Action" <%= "Action".equals(genre) ? "selected" : "" %>>Action</option>
                        <option value="Comedy" <%= "Comedy".equals(genre) ? "selected" : "" %>>Comedy</option>
                        <option value="Crime" <%= "Crime".equals(genre) ? "selected" : "" %>>Crime</option>
                        <option value="Drama" <%= "Drama".equals(genre) ? "selected" : "" %>>Drama</option>
                        <option value="Fantasy" <%= "Fantasy".equals(genre) ? "selected" : "" %>>Fantasy</option>
                        <option value="Horror" <%= "Horror".equals(genre) ? "selected" : "" %>>Horror</option>
```

```
                <option value="Romance" <%= "Romance".equals(genre) ? "selected" : "" %>>Romance</option>
                <option value="Sci-Fi" <%= "Sci-Fi".equals(genre) ? "selected" : "" %>>Sci-Fi</option>
                <option value="Thriller" <%= "Thriller".equals(genre) ? "selected" : "" %>>Thriller</option>
            </select>
        </td>
    </tr>
    <tr>
        <td><label for="releaseYear">Release Year:</label></td>
        <td><input type="number" id="releaseYear" name="releaseYear" min="1888" max="2030" required value="<%=
releaseYear %>"></td>
    </tr>
    <tr>
        <td><label for="rating">MPAA Rating:</label></td>
        <td>
            <select id="rating" name="rating" required>
                <option value="G" <%= "G".equals(rating) ? "selected" : "" %>>G</option>
                <option value="PG" <%= "PG".equals(rating) ? "selected" : "" %>>PG</option>
                <option value="PG-13" <%= "PG-13".equals(rating) ? "selected" : "" %>>PG-13</option>
                <option value="R" <%= "R".equals(rating) ? "selected" : "" %>>R</option>
                <option value="NC-17" <%= "NC-17".equals(rating) ? "selected" : "" %>>NC-17</option>
            </select>
        </td>
    </tr>
    <tr>
        <td><label for="director">Director:</label></td>
        <td><input type="text" id="director" name="director" required value="<%= director %>"></td>
    </tr>
  </table>

  <div class="form-actions">
    <button type="submit" class="btn-submit">Update Movie</button>
    <a href="movieDisplay.jsp" class="btn-reset">Cancel</a>
  </div>
</form>

<h3>Field Descriptions</h3>
<ul class="field-descriptions">
  <li><strong>Movie Title:</strong> The name of the movie</li>
  <li><strong>Genre:</strong> The category or type of movie</li>
  <li><strong>Release Year:</strong> The year the movie was released</li>
  <li><strong>MPAA Rating:</strong> Motion picture rating (G, PG, PG-13, R, NC-17)</li>
```

```
            <li><strong>Director:</strong> The director of the movie</li>
        </ul>

        <div class="form-actions">
            <a href="movieDisplay.jsp" class="btn-reset">View All Movies</a>
            <a href="index.jsp" class="btn-reset">Back to Course Index</a>
        </div>
<%
        }
%>
    </div>

    <footer>
        <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
</body>
</html>
```

movieSelection.jsp

```
<%--
    CSD430 Server-Side Development

    Module 8 Project Part 3: CRUD-UPDATE Operations
    Author: Brittaney Perry-Morgan

    Description: Selection page with dropdown to choose a movie to edit.
    The dropdown displays all movie IDs from the database.

    movieSelection.jsp - Movie Selection Page
--%>
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="csd430.MovieBean" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Select Movie to Edit - CSD430</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
```

```
        <h1>Module 8 Project Part 3</h1>
        <h2>Movie Database - <em>UPDATE Operation</em></h2>
        <p class="description">Select a movie record to edit its details.</p>
    </div>

    <div class="container">
<%
        MovieBean movieBean = new MovieBean();
        boolean connected = movieBean.connect();

        if (!connected) {
%>
        <div class="error-message">
            <p>Database Connection Error:</p>
            <p><%= movieBean.getErrorMessage() %></p>
        </div>
        <div class="form-actions">
            <a href="index.jsp" class="btn-reset">Back to Course Index</a>
        </div>
<%
        } else {
            boolean success = movieBean.getAllMovies();

            if (!success) {
%>
        <div class="error-message">
            <p>Error retrieving movie data:</p>
            <p><%= movieBean.getErrorMessage() %></p>
        </div>
        <div class="form-actions">
            <a href="index.jsp" class="btn-reset">Back to Course Index</a>
        </div>
<%
            } else {
%>

        <h3>Select Movie to Edit</h3>
        <p class="description">
            Choose a movie from the dropdown below. The key field (Movie ID)
            will be displayed but cannot be edited.
        </p>
```

```
    <form action="movieEdit.jsp" method="get">
        <table class="form-table">
            <tr>
                <td><label for="movieId">Select Movie:</label></td>
                <td>
                    <select id="movieId" name="id" required>
                        <option value="">-- Select a Movie --</option>
<%
            while (movieBean.next()) {
%>
                        <option value="<%= movieBean.getMovieId() %>">
                            <%= movieBean.getMovieId() %> - <%= movieBean.getTitle() %> (<%= movieBean.getReleaseYear() %>)
                        </option>
<%
            }
            movieBean.disconnect();
%>
                    </select>
                </td>
            </tr>
        </table>

        <div class="form-actions">
            <button type="submit" class="btn-submit">Edit Selected Movie</button>
            <a href="movieDisplay.jsp" class="btn-reset">View All Movies</a>
        </div>
    </form>

    <h3>Field Descriptions</h3>
    <ul class="field-descriptions">
        <li><strong>Movie ID:</strong> Primary Key - displayed but cannot be modified</li>
        <li><strong>Title:</strong> The name of the movie (editable)</li>
        <li><strong>Genre:</strong> The category of the movie (editable)</li>
        <li><strong>Release Year:</strong> Year the movie was released (editable)</li>
        <li><strong>Rating:</strong> MPAA rating (editable)</li>
        <li><strong>Director:</strong> Director of the movie (editable)</li>
    </ul>

    <div class="form-actions">
        <a href="index.jsp" class="btn-reset">Back to Course Index</a>
```

```
        </div>
<%
            }
        }
%>
    </div>

    <footer>
        <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
</body>
</html>
```

movieUpdate.jsp

```
<%--
    CSD430 Server-Side Development

    Module 8 Project Part 3: CRUD-UPDATE Operations
    Author: Brittaney Perry-Morgan

    Description: Processes the movie update form submission.
    Validates input and updates the record in the database.

    movieUpdate.jsp - Update Processing Page
--%>
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="csd430.MovieBean" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Update Movie - CSD430</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 8 Project Part 3</h1>
        <h2>Movie Database - <em>UPDATE Operation</em></h2>
        <p class="description">Processing movie update...</p>
    </div>
```

```
  <div class="container">
<%
    // Set encoding
    request.setCharacterEncoding("UTF-8");

    // Retrieve form parameters
    String idParam = request.getParameter("movieId");
    String title = request.getParameter("title");
    String genre = request.getParameter("genre");
    String releaseYearStr = request.getParameter("releaseYear");
    String rating = request.getParameter("rating");
    String director = request.getParameter("director");

    // Validate movie ID
    int movieId = 0;
    try {
        movieId = Integer.parseInt(idParam);
    } catch (NumberFormatException e) {
%>
    <div class="error-message">
        <p>Invalid movie ID: <%= idParam %></p>
    </div>
    <div class="form-actions">
        <a href="movieDisplay.jsp" class="btn-submit">View All Movies</a>
        <a href="index.jsp" class="btn-reset">Back to Course Index</a>
    </div>
<%
        return;
    }

    // Validate required fields
    StringBuilder errorMsg = new StringBuilder();

    if (title == null || title.trim().isEmpty()) {
        errorMsg.append("Movie Title is required.<br>");
    }
    if (genre == null || genre.isEmpty()) {
        errorMsg.append("Genre is required.<br>");
    }
    if (releaseYearStr == null || releaseYearStr.isEmpty()) {
        errorMsg.append("Release Year is required.<br>");
```

```
    }
    if (rating == null || rating.isEmpty()) {
        errorMsg.append("Rating is required.<br>");
    }
    if (director == null || director.trim().isEmpty()) {
        errorMsg.append("Director is required.<br>");
    }

    if (errorMsg.length() > 0) {
%>
    <div class="error-message">
        <p><%= errorMsg.toString() %></p>
    </div>
    <div class="form-actions">
        <a href="movieEdit.jsp?id=<%= movieId %>" class="btn-submit">Edit Movie</a>
        <a href="movieDisplay.jsp" class="btn-reset">View All Movies</a>
    </div>
<%
        return;
    }

    // Parse release year
    int releaseYear = 0;
    try {
        releaseYear = Integer.parseInt(releaseYearStr);
    } catch (NumberFormatException e) {
%>
    <div class="error-message">
        <p>Invalid release year: <%= releaseYearStr %></p>
    </div>
    <div class="form-actions">
        <a href="movieEdit.jsp?id=<%= movieId %>" class="btn-submit">Edit Movie</a>
        <a href="movieDisplay.jsp" class="btn-reset">View All Movies</a>
    </div>
<%
        return;
    }

    // Update movie in database
    MovieBean movieBean = new MovieBean();
    boolean connected = movieBean.connect();
```

```
    if (!connected) {
%>
    <div class="error-message">
        <p>Database connection error: <%= movieBean.getErrorMessage() %></p>
    </div>
    <div class="form-actions">
        <a href="movieEdit.jsp?id=<%= movieId %>" class="btn-submit">Edit Movie</a>
        <a href="index.jsp" class="btn-reset">Back to Course Index</a>
    </div>
<%
        return;
    }

    boolean success = movieBean.updateMovie(movieId, title.trim(), genre, releaseYear, rating, director.trim());
    movieBean.disconnect();

    if (success) {
%>
    <div class="success-message">
        <p>Movie updated successfully!</p>
    </div>

    <h3>Updated Movie Details</h3>
    <table>
        <thead>
            <tr>
                <th>Field</th>
                <th>Value</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td><strong>Movie ID</strong></td>
                <td><%= movieId %></td>
            </tr>
            <tr>
                <td><strong>Title</strong></td>
                <td><%= title.trim() %></td>
            </tr>
            <tr>
```

```
            <td><strong>Genre</strong></td>
            <td><%= genre %></td>
        </tr>
        <tr>
            <td><strong>Release Year</strong></td>
            <td><%= releaseYear %></td>
        </tr>
        <tr>
            <td><strong>Rating</strong></td>
            <td><%= rating %></td>
        </tr>
        <tr>
            <td><strong>Director</strong></td>
            <td><%= director.trim() %></td>
        </tr>
    </tbody>
</table>

<h3>Field Descriptions</h3>
<ul class="field-descriptions">
    <li><strong>Movie ID:</strong> Unique identifier (Primary Key, cannot be changed)</li>
    <li><strong>Title:</strong> The name of the movie</li>
    <li><strong>Genre:</strong> The category of the movie</li>
    <li><strong>Release Year:</strong> Year the movie was released</li>
    <li><strong>Rating:</strong> MPAA rating</li>
    <li><strong>Director:</strong> Director of the movie</li>
</ul>

<div class="form-actions">
    <a href="movieDisplay.jsp" class="btn-submit">View All Movies</a>
    <a href="/module_07_project_2/movieForm.jsp" class="btn-reset">Add New Movie</a>
    <a href="movieEdit.jsp?id=<%= movieId %>" class="btn-reset">Edit Again</a>
</div>
<%
    } else {
%>
<div class="error-message">
    <p>Failed to update movie: <%= movieBean.getErrorMessage() %></p>
</div>
<div class="form-actions">
    <a href="movieEdit.jsp?id=<%= movieId %>" class="btn-submit">Edit Movie</a>
```

```
            <a href="movieDisplay.jsp" class="btn-reset">View All Movies</a>
        </div>
<%
        }
%>
    </div>

    <footer>
        <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
</body>
</html>
```

deliverables

MODULE 8 PROJECT PART 3

JDBC & JAVABEANS - CRUD-UPDATE OPERATIONS

Author: Brittaney Perry-Morgan

MOVIE DATABASE CRUD OPERATIONS

DATABASE INFORMATION

Database Name: CSD430

Table Name: brittaney_movies_data

Username: student1

Topic: Movies

Fields: 6 (movie_id, title, genre, release_year, rating, director)

NAVIGATION - UPDATE OPERATIONS

Edit Movie

Select a movie from dropdown to edit

View All Movies

Display all movie records with edit links

Add New Movie

Create a new movie record (Module 7 CREATE)

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 7 PROJECT PART 2

MOVIE DATABASE - CREATE OPERATION

Add a new movie record to the database.

ADD NEW MOVIE

Movie Title:	Enter movie title
Genre:	-- Select Genre --
Release Year:	e.g., 2024
MPAA Rating:	-- Select Rating --
Director:	Enter director name

Add Movie **Clear Form**

FIELD DESCRIPTIONS

- **Movie Title:** The name of the movie
- **Genre:** The category or type of movie (Action, Comedy, etc.)
- **Release Year:** The year the movie was released
- **MPAA Rating:** Motion picture rating (G, PG, PG-13, R, NC-17)
- **Director:** The director of the movie

View All Movies **Back to Course Index**

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 8 PROJECT PART 3

MOVIE DATABASE - UPDATE OPERATION

Select a movie to edit its details.

Total Records: 10

MOVIE ID	TITLE	GENRE	RELEASE YEAR	RATING	DIRECTOR	ACTIONS
1	The Shawshank Redemption	Drama	1994	R	Frank Darabont	Edit
2	The Godfather	Crime	1972	R	Francis Ford Coppola	Edit
3	The Dark Knight	Action	2008	PG-13	Christopher Nolan	Edit
4	Pulp Fiction	Crime	1994	R	Quentin Tarantino	Edit
5	The Lord of the Rings: The Return of the King	Fantasy	2003	PG-13	Peter Jackson	Edit
6	Forrest Gump	Drama	1994	PG-13	Robert Zemeckis	Edit
7	Inception	Sci-Fi	2010	PG-13	Christopher Nolan	Edit
8	The Matrix	Sci-Fi	1999	R	The Wachowskis	Edit
9	Titanic	Romance	1997	PG-13	James Cameron	Edit
10	The Silence of the Lambs	Thriller	1991	R	Jonathan Demme	Edit

FIELD DESCRIPTIONS

- **Movie ID:** Unique identifier (Primary Key)
- **Title:** The name of the movie
- **Genre:** The category of the movie
- **Release Year:** Year the movie was released
- **Rating:** MPAA rating
- **Director:** Director of the movie

Add New Movie | Back to Course Index

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 8 PROJECT PART 3

MOVIE DATABASE - UPDATE OPERATION

Edit the movie details below.

EDIT MOVIE

Movie ID:	1
Movie Title:	The Shawshank Redemption
Genre:	Drama
Release Year:	1994
MPAA Rating:	R
Director:	Frank Darabont

Update Movie **Cancel**

FIELD DESCRIPTIONS

- **Movie Title:** The name of the movie
- **Genre:** The category or type of movie
- **Release Year:** The year the movie was released
- **MPAA Rating:** Motion picture rating (G, PG, PG-13, R, NC-17)
- **Director:** The director of the movie

View All Movies **Back to Course Index**

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 8 PROJECT PART 3

MOVIE DATABASE - UPDATE OPERATION

Edit the movie details below.

EDIT MOVIE

Movie ID:	1
Movie Title:	The Shawshank Redemption
Genre:	Action
Release Year:	1994
MPAA Rating:	R
Director:	Frank Darabont

Update Movie **Cancel**

FIELD DESCRIPTIONS

- **Movie Title:** The name of the movie
- **Genre:** The category or type of movie
- **Release Year:** The year the movie was released
- **MPAA Rating:** Motion picture rating (G, PG, PG-13, R, NC-17)
- **Director:** The director of the movie

View All Movies **Back to Course Index**

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 8 PROJECT PART 3

MOVIE DATABASE - UPDATE OPERATION

Processing movie update...

Movie updated successfully!

UPDATED MOVIE DETAILS

FIELD	VALUE
Movie ID	1
Title	The Shawshank Redemption
Genre	Action
Release Year	1994
Rating	R
Director	Frank Darabont

FIELD DESCRIPTIONS

- **Movie ID:** Unique identifier (Primary Key, cannot be changed)
- **Title:** The name of the movie
- **Genre:** The category of the movie
- **Release Year:** Year the movie was released
- **Rating:** MPAA rating
- **Director:** Director of the movie

View All Movies | Add New Movie | Edit Again

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development

MODULE 8 PROJECT PART 3

MOVIE DATABASE - UPDATE OPERATION

Processing movie update...

Movie updated successfully!

UPDATED MOVIE DETAILS

FIELD	VALUE
Movie ID	1
Title	The Shawshank Redemption
Genre	Action
Release Year	1994
Rating	R
Director	Frank Darabont

FIELD DESCRIPTIONS

- **Movie ID:** Unique identifier (Primary Key, cannot be changed)
- **Title:** The name of the movie
- **Genre:** The category of the movie
- **Release Year:** Year the movie was released
- **Rating:** MPAA rating
- **Director:** Director of the movie

View All Movies | Add New Movie | Edit Again

© 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development